

Tunneling through a barrier with the phase–amplitude method

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ABSTRACT

A previous study (Rawitscher, 2015) of the solution of Milne's non linear equation for the phase and amplitude of a one-dimensional wave function is extended to the case where the incident energy is less than the potential (Barrier region). The numerical method again consists in implementing a spectral expansion of the amplitude in terms of a number of Chebyshev polynomials. The method is applied to a Morse-type potential, for energies in a resonance region, and for one energy below the resonance region, and a strong repulsive Coulomb potential. The results are compared with highly accurate direct solutions of the Schrödinger equation, and were found to be accurate to $1:10^{-6}$.

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1. Introduction

The tunneling of a wave function through a potential barrier is a well known phenomenon in quantum mechanics. It has played an important role in the interpretation of various physical phenomena, such as the decay of a nucleus through α -particle emission, the fission [1] or fusion [2,3] of a heavy nucleus at energies below the Coulomb barrier, optical model scattering calculations [4–6], in the physics of ultra-low temperature atoms [7], and may have applications in other fields of physics [8] as well. Since the 1930s the Wentzel, Kramers, Brillouin (WKB) approximation provided a very useful tool in the calculation of barrier penetration probabilities. A good pedagogical description of the WKB approximation with several good references can be found in the book by Griffith [9]. Since the WKB expression contains factors $\exp(\pm \int^r \kappa(r') dr')$, where κ is the local wave number, which are exponentially sensitive to the errors of the WKB, it would be very helpful to find improvements to this approximation. Many such improvements have been obtained [10], and it is the purpose of the present note to present yet another improvement based on the phase–amplitude (Ph–A) description of the solution of a one-variable Schrödinger equation.

Milne's equations for the Ph–A description [11] have a non-linear term which presents difficulties for the numerical solution. Seaton and Peach [12] have presented an iterative solution that proved to be very accurate once a spectral expansion [13,14] for the calculation of the amplitude was introduced [15]. However

the latter formulation was up to now applied only to attractive potentials, in the region where the local wave number is real, in contrast to a barrier region, where the local wave number is purely imaginary. It is the purpose of the present paper to remedy that lack, and show that the appropriate modification of the Ph–A method also works for the radial region that is forbidden classically. However, the restrictions for the validity of the WKB approximation are valid also for the method presented here, i.e., the method is accurate only in a region sufficiently far away from the turning points such that the variations of the potential in a distance of the local wave length are small compared to the potential itself [9].

In Section 2 the Ph–A formulas pertinent to the barrier region will be derived, in Section 3 the iterative solution method will be described, in Section 4 the connection formulas across the turning points will be presented, in Section 5 a numerical application to a Morse-type potential will be described for incident energies near and below the resonance region. Section 6 contains results for the long range Coulomb potential, Section 7 contains some numerical considerations, and Section 8 contains the summary and conclusions.

The resonance properties for the Morse potential case have been investigated previously [16], and a comparison of the accuracy of various computational methods that calculate the scattering phase shifts in the resonance region was presented in Ref. [17]. The accuracy analyses was possible, because the phase shifts are known analytically for the Morse potential case. The resonance region presents a challenging test of accuracy, because some of the wave functions do decrease with distance r in that region, while the errors of the computational method introduce increasing contributions in the barrier region. It was shown in

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Fig. 2 of Ref. [17] that using a sixth order Numerov method for the solution of the Schrödinger equation the results were six orders of magnitude less accurate than that of a solution based on a spectral integral equation method denoted as S-IEM [18]. The latter method will also be used in the present investigation as a comparison benchmark result for the numerical examples.

2. The equations

The one-dimensional radial Schrödinger equation for a partial wave function ψ is written as

$$d^2\psi/dr^2 + k^2\psi = V\psi \quad (1)$$

where V is the spherically symmetric potential, that includes the angular momentum contribution $L(L+1)/r^2$, and k is the wave number. The corresponding energy of the incident particle is k^2 , which is assumed to be positive. In the present notation the factor $2m/\hbar^2$ has already been multiplied into quantities given in energy units, such that the resulting values for V and E have units of inverse length squared, while k is given in units of inverse length, and r is given in units of length. The units will however not be indicated in the graphs below. In the barrier region, where $V > E$, the local wave number κ is given by

$$\kappa(r) = \sqrt{\tilde{w}(r)}, \quad (2)$$

where

$$\tilde{w}(r) = V(r) - k^2 > 0. \quad (3)$$

If one inserts into Eq. (1) the ansatz

$$\psi^{(-)}(r) = y^{(-)}(r) \exp(-\Phi^{(-)}(r)) \quad (4)$$

or

$$\psi^{(+)}(r) = y^{(+)}(r) \exp(\Phi^{(+)}(r)) \quad (5)$$

or

$$\psi(r) = A\psi^{(-)}(r) + B\psi^{(+)}(r)$$

one finds that

$$y^{(+)} = y^{(-)} = y \quad (6)$$

and

$$\Phi^{(+)} = \Phi^{(-)} = \Phi(r) \quad (7)$$

where the common amplitude y obeys the equation

$$\frac{d^2}{dr^2}y - \tilde{w}y = -\frac{k^2}{y^3} \quad (8)$$

and the common phase Φ is given by the simple quadrature

$$\Phi(r) = \int_a^r \frac{k}{y^2(r')} dr'. \quad (9)$$

The two turning points at the extremities of the barrier region are T_1 and T_2 , and the region suitable for the Ph-A method is located in $[a, b]$, which is contained between T_1 and T_2 . (Hence $T_1 < a \leq r \leq b < T_2$.) The complete Ph-A wave function in $[a, b]$ is given by

$$\psi(r) = y(r) [A e^{-\Phi(r)} + B e^{+\Phi(r)}]; \quad a \leq r \leq b, \quad (10)$$

where the coefficients A and B are determined by the connection formulas across the turning points. The validity of Eqs. (6)–(10) can be verified by inserting Eq. (10) into Eq. (1), and setting to zero each of the terms multiplied respectively the factors A or B .

Once the phase is defined according to Eq. (9), then the relationship between phase Φ and amplitude y is determined uniquely, but other relationships are also possible [19], as is the

case for the Calogero's Ph-A formalism, described in Appendix A of Ref. [15]. The region of validity of the WKB approximation requires that the local wave length $\lambda(r) = 2\pi k/[E - V(r)]$ changes little in the distance of the local wave length, i. e., that

$$\frac{\Delta\lambda}{\lambda} = \frac{2\pi k}{(V-E)^2} \frac{dV}{dr} \ll 1. \quad (11)$$

However the conditions of applicability of the Ph-A method are more stringent, because the convergence of the iterations depends on the smallness of the quantity $(d^2y/d^2r)/y$ compared to the potential V .

3. Iterative solution

In this section the iterative method of Seaton and Peach [12] is extended to the barrier region, where $\tilde{w} = V - k^2 > 0$. By re-writing Eq. (8) and taking square roots, one obtains

$$\frac{k}{y_{n+1}^2} = (\tilde{D}_n + \tilde{w})^{1/2}, \quad n = 0, 1, 2.. \quad (12)$$

where $\tilde{w}$ is defined in Eq. (3), where

$$\tilde{D}_n = -\frac{d^2y_n/dr^2}{y_n} \quad (13)$$

and where $D_0 = 0$. The resulting value of y_1 is identical to the WKB approximation

$$y_1 = y_{\text{WKB}} = (V/k^2 - 1)^{-1/4}, \quad a \leq r \leq b \quad (14)$$

and hence the phase and the wave function (10) becomes identical to their WKB values. Since there would be a numerical loss of accuracy when calculating the second order derivative of y , it is preferable to obtain $\tilde{D}_n$ by a recursion relation which in the present case takes the form

$$\tilde{D}_{n+1} = -\frac{5}{16} \frac{(\tilde{D}_n' + \tilde{w}')^2}{(\tilde{D}_n + \tilde{w})^2} + \frac{1}{4} \frac{(\tilde{D}_n'' + \tilde{w}'')}{(\tilde{D}_n + \tilde{w})}, \quad n = 0, 1, 2, \dots, \quad (15)$$

where “primes” denote derivatives with respect to r . For $n = 0$, one has $\tilde{D}_0 = 0$, and hence all its derivatives are zero. The derivatives of $\tilde{w}$ are equal to the derivatives of V , which can be calculated analytically if the analytic expression for V is known, as is the case in the numerical example below. The advantage of obtaining $\tilde{D}_{n+1}$ by means of Eq. (15) instead of by calculating the second derivative of y_{n+1} directly is that the quantity $\tilde{D}_{n+1}$ and its derivatives are small compared to $\tilde{w}$ and its derivatives, and hence the effect of the errors of the Chebyshev expansion of $\tilde{D}_{n+1}$ and its derivatives becomes reduced. A numerical comparison of the potential V and the quantity D is presented in connection with an application of the Ph-A method to a Coulomb potential.

There are two methods for obtaining $\tilde{D}$. One consists by inserting the values of $\tilde{D}_{n-1}$ and its derivatives into Eq. (15) in order to obtain $\tilde{D}_n$. Next, the derivatives of $\tilde{D}_n$, calculated numerically, are inserted into the right hand side of Eq. (15) and the iteration for $\tilde{D}_n$ continues until the iteration converges for n_{max} . By inserting the result into Eq. (12) one obtains the values of $y_{n_{\text{max}}}$ for all support points, hence the phase for all support points can be obtained from the quadrature indicated in Eq. (9) from which the functions $\psi^{(\pm)}$ can be obtained. This procedure is especially economical if a spectral expansion of all the functions in terms of Chebyshev polynomials is implemented, as described below. The coefficients A and B required for the full wave function in Eq. (10) are obtained by a connection formula across the left turning point, as described below.

A second iterative method for obtaining $\tilde{D}$ is as follows: One assumes fixed values for the first and second derivatives of $\tilde{D}$, denoted as $\tilde{D}'_M$ and $\tilde{D}''_M$ and inserts them into the numerators of Eq. (15) that can be written as

$$\tilde{z}_{n+1} = -\frac{5}{16} \frac{(\tilde{D}'_M + \tilde{w}')^2}{(\tilde{z}_n)^2} + \frac{1}{4} \frac{(\tilde{D}''_M + \tilde{w}'')}{(\tilde{z}_n)} + \tilde{w}, \quad n = 0, 1, 2, \quad (16)$$

where

$$\tilde{z}_n = \tilde{D}_n + \tilde{w}. \quad (17)$$

After convergence at $n = n_{\max}$, the value of $\tilde{z}_{n_{\max}+1}$ can be set equal to $\tilde{z}_{n_{\max}} = \tilde{z}_M$, and Eq. (16) can be transformed into the cubic equation

$$\tilde{z}_M^3 = -\frac{5}{16} (\tilde{D}'_M + \tilde{w}')^2 + \frac{1}{4} (\tilde{D}''_M + \tilde{w}'') (\tilde{z}_M) + \tilde{w} (\tilde{z}_M)^2. \quad (18)$$

The solution of Eq. (18) does not require iterations but can be solved by standard algebraic means [20], as explicitly described in Ref. [15]. The derivatives of the resulting values of $\tilde{z}_M$ are calculated numerically and are denoted by $\tilde{z}'_{M+1}$ and $\tilde{z}''_{M+1}$. They are inserted into Eq. (18) and the next value of $\tilde{z}$, denoted as $\tilde{z}_{M+1}$ is obtained. The two methods were compared in Ref. [15] for the case that $\tilde{D}'_M = \tilde{D}''_M = 0$, and excellent agreement was obtained. The values of y are obtained from Eq. (12), $y_M = (\tilde{z}_M/k^2)^{-1/4}$, and the calculation for the phase and wave function proceeds as described above. In the numerical examples given below, the iteration index M is denoted as n . Most of the results presented here are based on iterative method #2, although if method #1 does not converge, method #2 should not be used, as discussed further in the Appendix.

The numerical calculation consists in expanding the function $y(r)$ in the radial domain $[a \leq r \leq b]$ in terms of Chebyshev polynomials $T_n(x)$

$$y(r) = \sum_{n=1}^{N+1} a_n T_{n-1}(x). \quad (19)$$

The values of x are contained in the interval $[-1 < x < +1]$, and the corresponding values of r are related to the values of x by a linear mapping procedure. The calculation of the expansion coefficients a_n consists in considering the $N + 1$ support points ξ_n located at the zeros of T_{N+1} , [18], and solving the $(N + 1) \times (N + 1)$ matrix equations for the a_n 's. This procedure has been described in detail in Ref. [15], and has been given previously by Curtis and Clenshaw [21]. Once the coefficients a_n are known, Eq. (19) can be used to evaluate y at any continuous point in $[a, b]$.

4. Connection formulas

The wave function calculated numerically to the left side of the left turning point by some means other than the Ph-A method is denoted as $\psi_C(r)$ (the subscript C denotes "Conventional computation"). This wave function is continued numerically through the left turning point T_1 to two points r_1 and r_2 located in the barrier region where the Ph-A method is reliable, i.e. at $T_1 < a < r_1 < r_2 \ll b < T_2$. At these two points the Ph-A wave function is evaluated, with the coefficients A and B still to be determined, but the values of y and Φ known.

$$\psi_{Ph-A}(r_i) = y(r_i)[Ae^{-\Phi(r_i)} + Be^{\Phi(r_i)}], \quad i = 1, 2. \quad (20)$$

By equating the values of $\psi_{Ph-A}(r_i)$ to the values of $\psi_C(r_i)$, $i = 1, 2$, one obtains the matrix equation

$$\begin{pmatrix} y_1 e^{-\Phi_1} & y_1 e^{+\Phi_1} \\ y_2 e^{-\Phi_2} & y_2 e^{+\Phi_2} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} \psi_C(r_1) \\ \psi_C(r_2) \end{pmatrix} \quad (21)$$

that can be solved for the coefficients A and B . In the above $y_i = y(r_i)$ and $\Phi_i = \Phi(r_i)$, with $i = 1, 2$. In order to obtain reliable values for the coefficients A and B the condition number of the matrix in Eq. (21) has to be sufficiently small.

In order to connect the resulting Ph-A wave function across the right turning point, T_2 , to a conventional numerical wave function ψ_C , the methods described in section 5 of Ref. [15] can be adapted to the present situation. Two independent solutions of the Schrödinger equation can be calculated in the region across the right-hand turning point, and the appropriate linear combination can be obtained by the method described above.

In order to obtain the values of the wave functions $y_1 e^{\pm\Phi_1}$ and $y_2 e^{\pm\Phi_2}$ at the specified points r_1 and r_2 (or at r_3 and r_4 near the right-hand turning points), required for Eq. (21), an advantage of the spectral expansion (19) of the Ph-A representation is that the phase Φ and the amplitude y can be interpolated accurately to these points (or at any other points) in terms of the knowledge of these quantities at the spectral support points. Because both y and Φ vary more slowly with distance than the wave functions themselves the number of support points can be quite small, as is demonstrated in the numerical examples below.

5. Numerical example for the Morse potential

The Morse potential described in Ref. [16] is again chosen for the present application, because the two resonances that occur for this potential are known and well studied [16]. The accuracy analyses described in that study [16] and in the examples below were possible, because the phase shifts are known analytically for the Morse potential case.

Two cases will be examined for the incident momentum k : (1) In a region of resonance, and (2) In a region far from a resonance. The resonance region presents a challenging test of accuracy, because some of the wave functions do decrease with distance r in that region, while the errors of the computational method introduce increasing contributions in the barrier region.

The potential $V(r)$ is given by [16]

$$V(r) = V_0 e^{-(r-r_e)\alpha} [2 - e^{-(r-r_e)\alpha}], \quad (22)$$

with

$$V_0 = 6; \quad r_e = 4; \quad \alpha = 0.3. \quad (23)$$

Here V_0 is in units of inverse length squared, r_e in units of length, and α in units of inverse length. This potential, illustrated in Fig. 1, has a finite attractive valley near the origin, and a repulsive region beyond $r \simeq 2$. It supports one bound state, and two resonances at energies E_1 and E_2 , both indicated in the figure. The wave functions in the two resonance regions are shown in Figs. 2–4. The numbers in the legends label the momentum values k that span the respective resonances, as indicated in the captions to these figures.

The wave functions are obtained from the numerical S-IEM [18] solution of the wave equation in the radial region ($0 \leq r \leq 100$), and near $r = 100$ they are normalized such that the oscillation maxima are set to close to unity.

5.1. Resonance 1

The two turning points for the energy $E_1 = 1.1779$ in the vicinity of this resonance occur near $T_1 \simeq 1.9$ and $T_2 \simeq 11.7$.

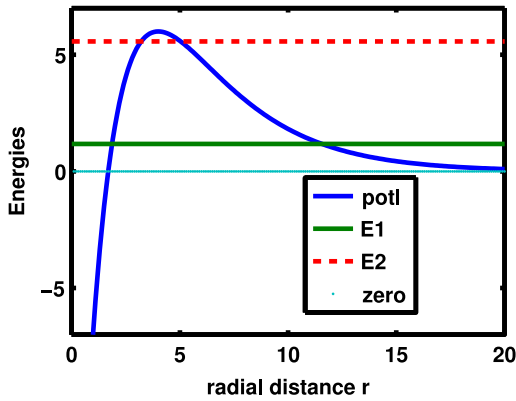


Fig. 1. The Morse potential is illustrated by the curved line. The horizontal lines indicate the location of the resonance energies, $E_1 = 1.1779$ and $E_2 = 5.57$, all in units of inverse length squared.

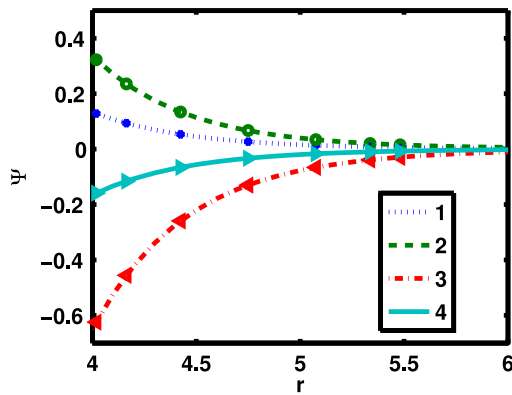


Fig. 2. The continuous lines illustrate the wave function results in the barrier region obtained by solving the Schrödinger equation for a Morse potential using the S-IEM method for $0 \leq r \leq 100$, for the various energies given by k^2 . Here the wave numbers k spanning the resonance region are given by $k = 1.08526787 + (n - 1) * 10^{-8}$, with $n = 1, \dots, 4$. The barrier region extends from $r = 2$ to $r = 12$. The discrete symbols represent the results of an independent Phase-Amplitude calculation described in the text.

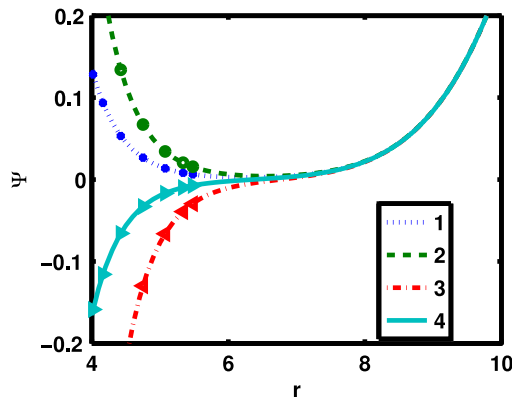


Fig. 3. The numerical solutions of the Schrödinger equation, are the same as in Fig. 2. In the present figure the radial region extends beyond the barrier, for the energies indicated in Fig. 2. The figure shows that these functions become almost identical at large distances, even though in the region near the origin they are quite different.

The region of convergence of the Ph-A approximation occurs in the radial interval approximately given by $r \in [3.5, 5.5]$, as is shown in Fig. 5. The values of the Ph-A amplitude y are obtained by the iterative method given by Eq. (18), with the initial value of the derivatives of D given by the WKB approximation. The rate of convergence of the iteration is displayed in Fig. 6. This figure shows

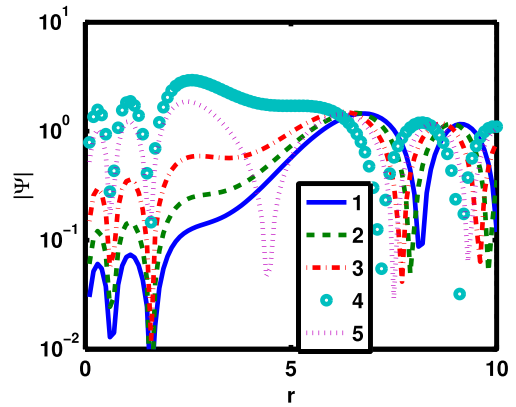


Fig. 4. Resonance 2 occurs near $k = 2.24$. The wave functions in that region have k -values given by $k = 2.24 + (n - 1) * 0.04$, with $n = 1, \dots, 5$. The barrier region extends from $r = 3$ to $r = 6$, and is too small for either the WKB or the present calculation to give meaningful results. The figure shows that for $k = 2.24$ and 2.28 ($n = 1$ and 2) the wave function inside the barrier region increases in absolute value while for $n = 4$ and 5 the wave function decreases, but have much larger absolute values in the barrier region.

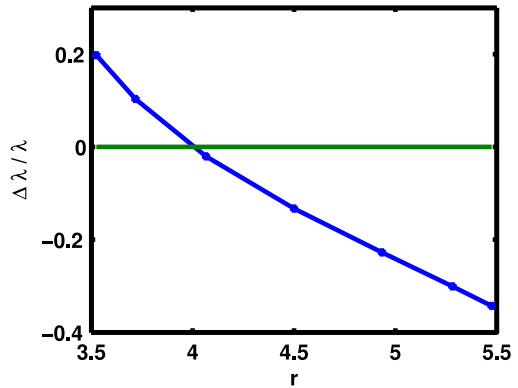


Fig. 5. The WKB test function $\Delta\lambda/\lambda$ defined in Eq. (11) for $k = 1.0855$, located in resonance #1. In the radial interval shown, the value of $|\Delta\lambda/\lambda|$ is acceptably smaller than unity.

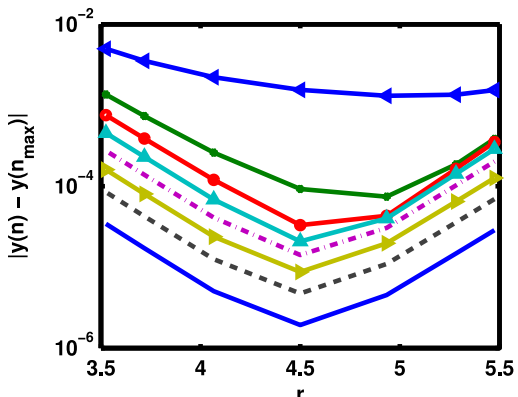


Fig. 6. The rate of convergence of the iterations for y described by Eq. (18), starting with the WKB approximation, for $k = 1.0855$, located in resonance #1. The maximum number of iterations is $n_{\max} = 8$, the top curve represents the absolute value of the difference between y_{WKB} and $y(n_{\max})$, and the successively lower curves represent $|y(n) - y(n_{\max})|$ for $n = 1, 2, \dots, 7$.

that the error of y for the WKB approximation is between 10^{-2} and 10^{-3} in the radial interval shown, while the error after the 8'th iteration is of the order of 10^{-5} .

The phase $\Phi(r)$ is displayed in Fig. 7. The amplitude y for $k = 1.0855$, is illustrated in Fig. 8. and the wave functions $\psi^{(+)}$ and $\psi^{(-)}$, defined as $y \exp(\pm i\Phi)$ are illustrated in Fig. 9.

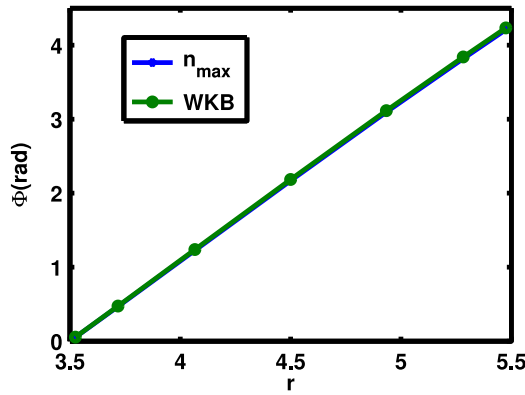


Fig. 7. The phase $\Phi(r)$, obtained from the amplitude y according to Eq. (9), for $k = 1.0855$, located in resonance #1. The difference between the WKB value and the result after 8 iterations ($n_{\max} = 8$) is not visible in this figure.

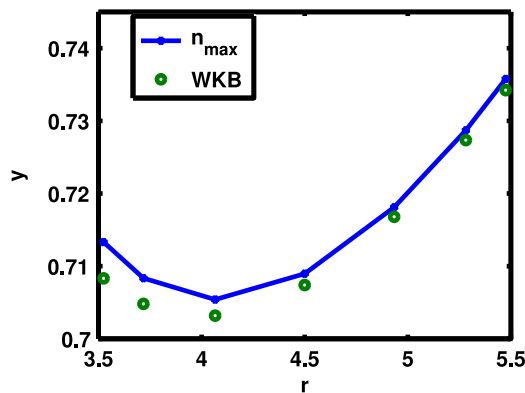


Fig. 8. The amplitude y for $k = 1.0855$, in the barrier region. The result after 8 iterations clearly differs from the WKB approximation.

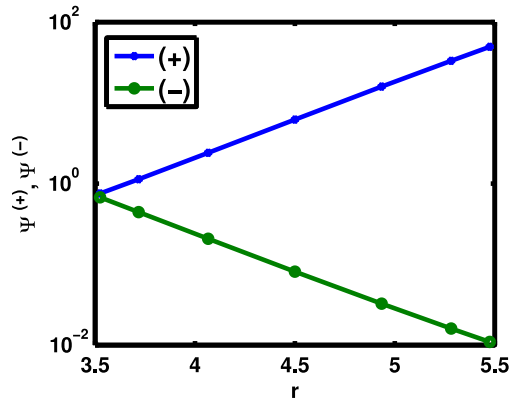


Fig. 9. The wave functions $\psi^{(\pm)} = y \exp(\pm\Phi)$ as a function of the location r in the barrier region for $k = 1.0855$, after 8 iterations.

The calculation of the Ph-A wave function, defined in Eq. (10) requires the calculation of the coefficients A and B , which in turn, according to Eq. (21), depend on the wave function ψ_C obtained at two (arbitrary) points r_1 and r_2 located inside the region $[a, b]$ near the left end. In the present case ψ_C is calculated by means of the S-IEM method starting from the origin, and normalized asymptotically to have unit amplitude near $r = 100$, i.e., $\psi_C(r \rightarrow 100) = \sin(kr + \phi)$. The chosen values of r_1 and r_2 are 4.1 and 4.3, and the procedure is repeated for several values of k in the resonance region, given by

$$k(nk) = 1.085267870 + (nk - 1) \times 10^{-8},$$

$$nk = 1, 2, \dots, 10. \quad (24)$$

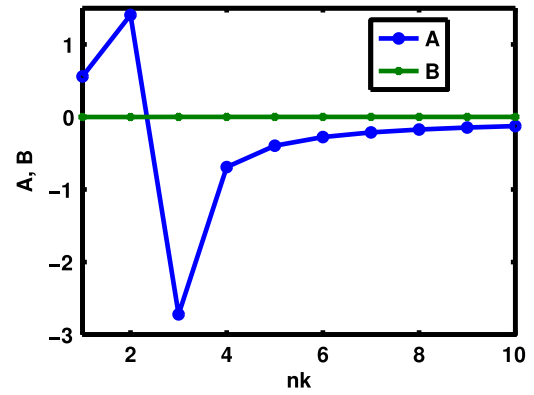


Fig. 10. The values of the coefficients A and B for the energies specified by the integers nk according to Eq. (24), for resonance #1.

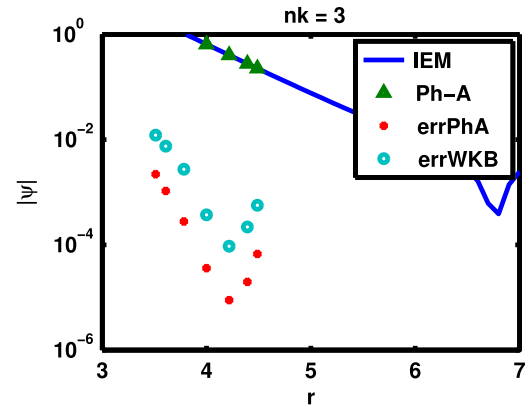


Fig. 11. The S-IEM and Ph-A wave functions are given by the solid line and the triangular symbols, respectively, for the wave number $k = 1.085267890$. The potential is described Eqs. (22) and (23). The errors of the WKB approximation and of the Ph-A result for the 8th iteration are given by the open and closed discrete symbols, respectively, at the positions of the support points for an expansion of y in terms of seven Chebyshev polynomials, in the interval $3.5 \leq r \leq 4.5$.

The resulting values of the Ph-A wave functions are illustrated by means of the solid symbols in Figs. 2 and 3. The agreement with the S-IEM wave functions, shown as continuous lines in these figures, is very satisfactory. However, even though the wave functions in the barrier region are very different from each other for the various k -values, asymptotically the wave functions are again very similar. The corresponding phase shifts ϕ differ from each other only in the 4th decimal place. This is not the case for the resonance #2 near the top of the barrier, Fig. 4, where for the various resonant energies the wave function differ strongly asymptotically from each other.

The values of the coefficients A and B for resonance 1 are displayed in Fig. 10 for the set of energies given by (24). It is clear that the resonance occurs for energies between $nk = 1$ and 4. The values of B are of the order or 10^{-5} , and are all positive, with the exception for $nk = 1$ and 2, for which they are negative. The error of the Ph-A wave function is displayed in Fig. 11. It can be seen from Fig. 11 that after 8 iterations the error of the Ph-A wave function is smaller than that of the WKB wave function by approximately one order of magnitude.

5.2. A non-resonant case

In order to illustrate a case of a wider barrier, the value of $k = 0.1$ is chosen. In this case the turning points are $T_1 \simeq 2$ and $T_2 \simeq 27$, but the region of convergence of the Ph-A iterations is contained approximately in the interval $[4 \leq r \leq 8]$, as is illustrated in Fig. 12. The number of Chebyshev functions in Eq. (19) is

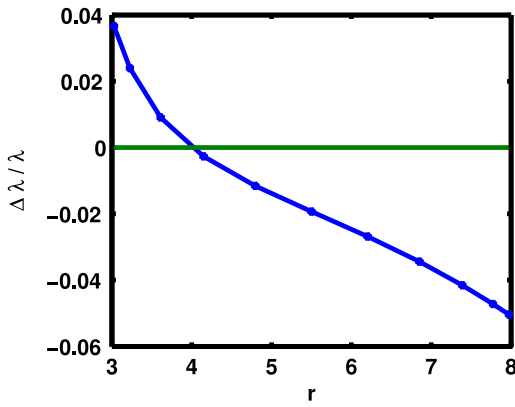


Fig. 12. Same as Fig. 5 for $k = 0.1$ for the Morse potential given by Eqs. (22) and (23).

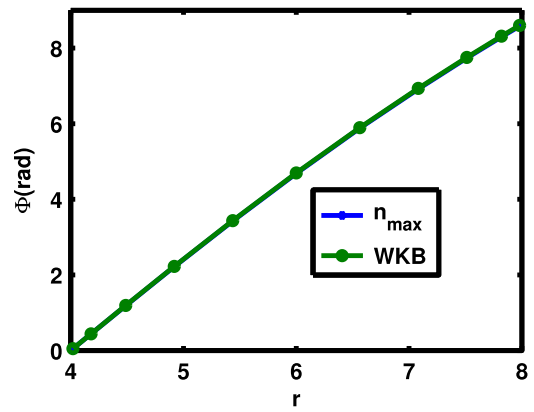


Fig. 15. Same as Fig. 14 for the phase Φ .

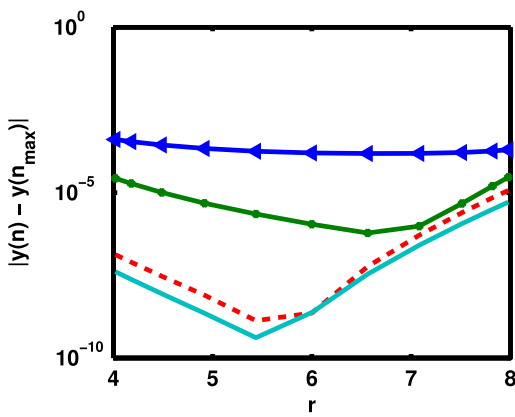


Fig. 13. The convergence of the iterations based on Eq. (18) for up to $n_{\max} = 8$ iterations is illustrated here for $k = 0.1$. The top line represents the absolute value of the difference between the WKB and $n_{\max}$ result. The next lower line represents the difference between the $n = 1$ and the $n_{\max}$ result, and the next two lines represent the same for $n = 6$ and 7 . The number of Chebyshev polynomials used in Eq. (19) is $N = 11$, and the radial interval is $[4, 8]$.

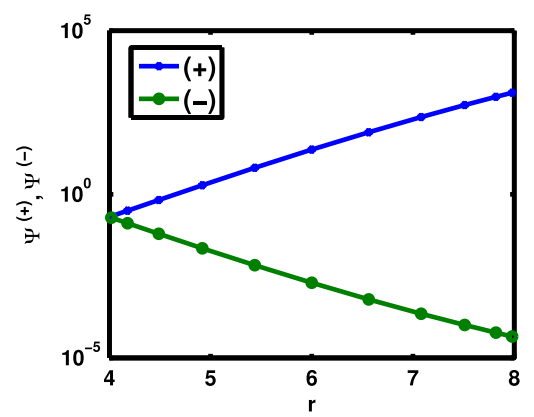


Fig. 16. The two wave functions $y \exp(\pm\Phi)$ in the barrier region, after the 8th iteration, for $k = 0.1$.

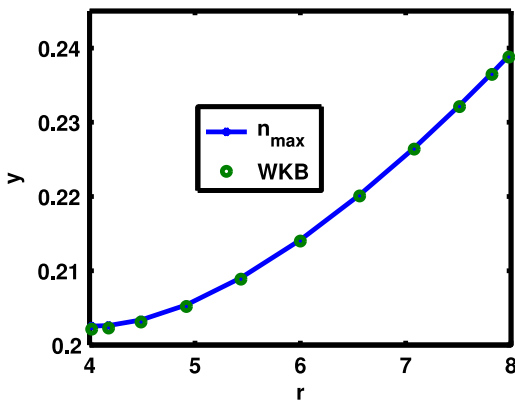


Fig. 14. The amplitude y as a function of distance in the barrier region. The number of Chebyshev polynomials is 11, $k = 0.1$. The results after the 8th iteration are visually indistinguishable from the WKB result. The symbols represent the Ph-A values at the support points.

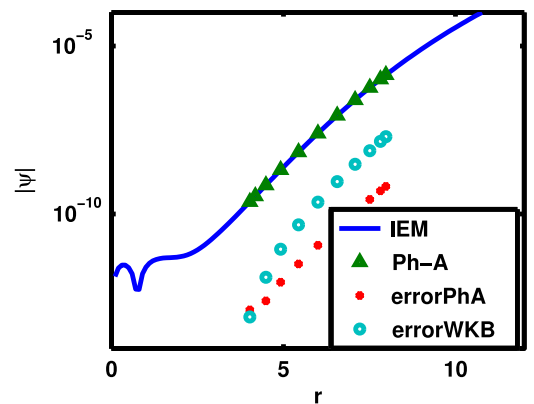


Fig. 17. The Morse potential wave function for the case $k = 0.1$ including the barrier region. The triangles illustrate the Ph-A results, obtained with 11 Chebyshev polynomials in the region $4 < r < 8$. The solid line represents the wave function obtained with the S-IEM method, and is normalized to unit amplitude near $r = 100$. The error points were obtained by comparing the values of the S-IEM wave functions with either the Ph-A values or the WKB values at the 11 Ph-A support points.

11, and the convergence up to 8 iterations performed with Eq. (18) is illustrated in Fig. 13. The dependence of the amplitude y on r is illustrated in Fig. 14, and the corresponding phase $\Phi(r)$ is shown in Fig. 15. The wave functions $\psi^{(\pm)}$ are illustrated in Fig. 16.

Since for this $k = 0.1$ case the local wave number is larger than for the resonance case #1, for which $k \simeq 1.0853$, the amplitude y is smaller (since the barrier is deeper), the phase Φ is larger, and

the functions $\psi^{(\pm)}$ change more steeply with distance than for the $k \simeq 1.0853$ case, as can be easily understood from the WKB approximation. The absolute values of the Ph-A wave function points are displayed by means of the discrete triangles in Fig. 17, and the errors of these values are displayed by means of the circles.

6. Numerical examples for the $1/r$ Coulomb case

The conventional equation for the wave function $\chi(\rho)$ describing the point Coulomb case is

$$\frac{d^2\chi}{d\rho^2} + \left[1 - \frac{2\eta}{\rho} - \frac{L(L+1)}{\rho^2}\right]\chi = 0, \quad (25)$$

where ρ describes the radial distance, and η describes the strength of the Coulomb interaction. Both quantities are dimensionless. The corresponding potential in Eq. (1) is given by

$$V(r) = \frac{Ze^2}{r} \frac{2m}{\hbar^2} = \frac{\bar{Z}}{r}, \quad (26)$$

where $\bar{Z}$ has units of inverse length and is related to 2η according to

$$\bar{Z} = 2\eta k \quad (27)$$

where

$$2\eta = Z \frac{e^2}{\hbar c} \left(\frac{mc^2}{2E}\right)^{1/2}. \quad (28)$$

In the above m is the mass and E the energy of the incident particle, Z the number of protons in the nucleus, e the charge of a proton, $\hbar$ is Planck's constant, and $\rho = kr$. For the attractive case, Z and η are both negative. Asymptotically a distorted Coulomb wave function approaches

$$\chi(\rho) \rightarrow \sin(\rho - \eta \ln(2\rho) - L\pi/2 + \theta_\infty) \quad (29)$$

where θ_∞ is the phase shift, whose value depends on how the potential differs from the point Coulomb potential. The position dependent phase is defined as

$$\theta(\rho) = \Phi(\rho) - [\rho - \eta \ln(2\rho) - L\pi/2], \quad (30)$$

which asymptotically approaches the phase shift θ_∞ , and where Φ is the phase of the Ph-A method.

In order to mitigate the singularities at the origin of r , a “rounding” procedure is introduced for the numerical examples. This procedure replaces r by $\mathcal{R}(r)$ by means of the analytic transformation

$$\mathcal{R}(r) = r/[1 - \exp(-r/t)], \quad (31)$$

such that $\mathcal{R}(r)$ approaches a constant in the limit $r \rightarrow 0$ but does not alter the potential at distances much larger than the rounding parameter t . In the Schrödinger equation (1), Eq. (26) is replaced by

$$V(r) = \frac{Ze^2}{\mathcal{R}(r)} \frac{2m}{\hbar^2} = \frac{\bar{Z}}{\mathcal{R}(r)}, \quad (32)$$

and in Eq. (25) $2\eta/\rho$ is replaced by $2\eta/\tilde{\rho}$, where $\tilde{\rho} = k\mathcal{R}(r)$. In this case the differential equation to be solved is

$$\frac{d^2\chi}{d\rho^2} + \left[1 - \frac{2\eta}{\tilde{\rho}} - \frac{L(L+1)}{\rho^2}\right]\chi = 0, \quad (33)$$

where $\rho = kr$, $\tilde{\rho} = k\mathcal{R}(r)$ and $\eta = \bar{Z}/(2k)$ are dimensionless.

Many methods have been developed for obtaining the point-Coulomb wave functions, several of which are listed in Chapter 33 of Ref. [20], especially in Section 33.23 of that reference. An excellent review paper is given by Seaton [22]. Additional methods have also been proposed [23] for distorted Coulomb functions.

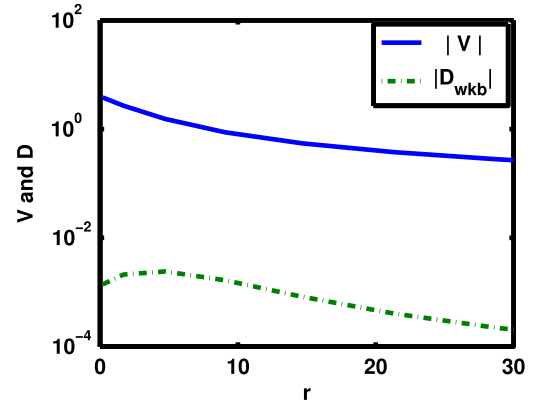


Fig. 18. The figure compares the potential V with the quantity $\tilde{D}$ for the rounded Coulomb potential case described in this section. It shows that $|D|$ is several orders of magnitude smaller than $|V|$.

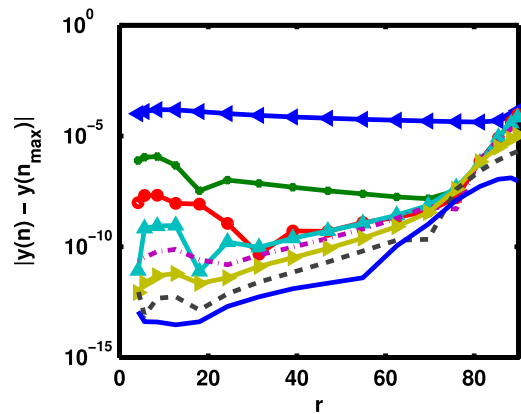


Fig. 19. The convergence of the iterations of the amplitude y for the rounded Coulomb case with $Z = 8$ and $k = 0.1$, with $4 \leq r \leq 90$, using 17 Chebyshev expansion polynomials. The corresponding value of η is 40. The iterations are based on Eqs. (16) and (18). The top curve with the triangle symbols pointing to the left represents the WKB approximation, and the subsequent curves in descending order correspond to, respectively, $n = 1, 2, \dots, 7$.

6.1. Numerical example #1

In this example $t = 2$, $\bar{Z} = 8$ ($\bar{Z}$ has units of inverse length), V in $V(r)$ has units of inverse length squared, $k = 0.1$ (in units of inverse length), and $\eta = 40$. The convergence of the iterations of Eqs. (15) or (16) requires that $D \ll V$. This is indeed the case for the present example, as illustrated in Fig. 18.

The convergence of successive values of y_n is shown by a plot of $|y_n - y_{n_{\max}}|$, $n = 1, 2, \dots, n_{\max} - 1$ in Fig. 19. After $n = 8$ iterations, the resulting values of the amplitude and phase are illustrated in Figs. 20 and 21. The resulting functions $\psi^{(+),(-)}$ are illustrated in Fig. 22. As this figure shows, these wave functions change by many orders of magnitude, which, for the case of $\psi^{(-)}$, would present substantial accuracy problems.

6.2. Numerical example #2

The input parameters are the same as in Example #1, with the exception that now $\bar{Z} = 4$. The corresponding value of η is 20, which is the largest value for which a table of the regular Coulomb wave functions $F_0(\eta, \rho)$ values can be found in Table 14.1 in Ref. [24] for comparison purposes. In the Ph-A solution of Eq. (1) the range of r values is $(0 \leq r \leq 60)$ and the iterations are calculated by means of Eqs. (16) and (18). The resulting function $\psi^{(+)}(r) = y(r) e^{+\Phi(r)}$ is proportional to the regular (distorted) Coulomb function, but since it has not been continued beyond

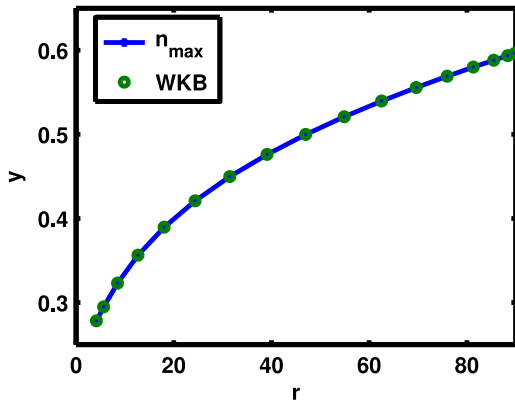


Fig. 20. The amplitude y of the solution of Eq. (1) with the potential V given by Eq. (32), $Z = 8$, $k = 0.1$, $N = 16$, and $4 \leq r \leq 90$.

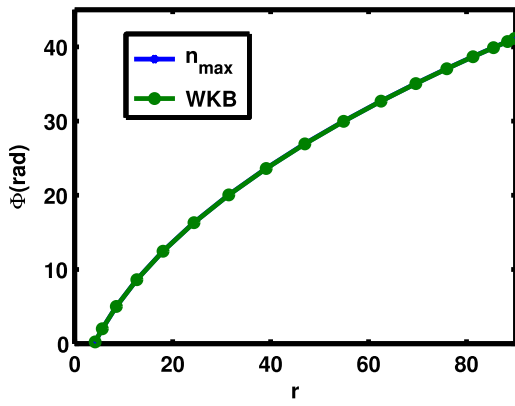


Fig. 21. The phase $\Phi(r)$ for the Coulomb parameters defined in Fig. 20.

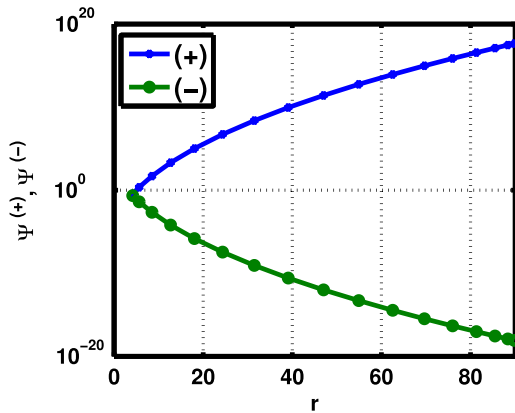


Fig. 22. The functions $\psi^{(+)}$ and $\psi^{(-)}$, solutions of Eq. (1) obtained with the Ph-A method, with parameters described in Fig. 20.

the outer turning point, the requirement that it approaches unit amplitude for $r \rightarrow \infty$ is not satisfied. Hence, $\psi^{(+)}(r)$ was normalized such that it agrees with $F_0(\eta, \rho)$ at $\rho = 3$, with the normalization factor $F_0(\eta, 3)/\psi^{(+)}(30) \simeq 8.72 \times 10^{-27}$. The comparison between the two functions is illustrated in Fig. 23. The good agreement shows that the Ph-A method gives reliable results for repulsive potentials as long ranged as the Coulomb potential.

7. Numerical considerations

The calculations are done with MATLAB on a desk PC using an Intel TM2 Quad, with a CPU Q 9950, a frequency of 2.83 GHz, and

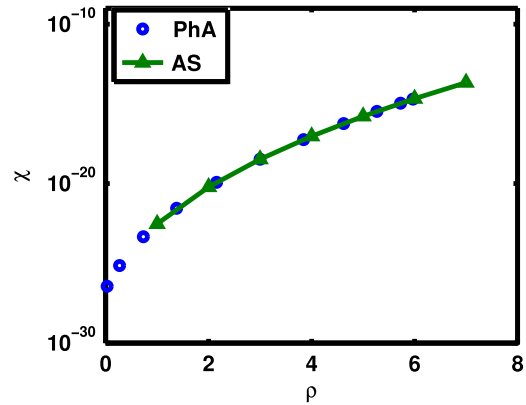


Fig. 23. Comparison of the $L = 0$, $\eta = 20$ regular Coulomb wave functions. The open circle results labeled with Ph A are obtained with the Phase-Amplitude method, and the results labeled with “~” are taken from Table 14.1 of Ref. [24]. The latter are connected by straight lines. The Ph-A results were normalized to the Table 14.1 of Ref. [24] at the point $\rho = 3.0$.

a RAM of 8 GB. The calculations use typically between $N + 1 = 7$ and 11 Chebyshev expansion polynomials.

Most of the numerical accuracy tests are done using the Morse-type potential displayed in Fig. 1. The comparison S-IEM wave functions are calculated [18] with an accuracy parameter of 10^{-11} . For the resonance #2 near the top of the barrier the Ph-A method could not be used, because the size of the barrier is too small. The accuracy and convergence of the Ph-A calculations for resonance #1 can be judged from Fig. 6. The accuracy for the non-resonant case for $k = 0.1$ can be judged from Fig. 13, according to which one could expect an accuracy of the order of 10^{-5} . The actual (relative) errors are shown in Fig. 17. Their values are $\simeq 5 \times 10^{-4}$ in reasonable agreement with the expectation. For the non-resonant case the conventional finite difference methods of calculating the wave function in the barrier region are perfectly acceptable, because the errors increase at the same rate as the wave function itself. This was shown by calculating the wave function with a 6th order Numerov method, whose initial two points were started from the S-IEM results near $r = 4$. The accuracy was consistently of order h^4 , as expected.

The Ph-A computational time for the $k = 0.1$ case, with 8 iterations, 11 Chebyshev polynomials, in the domain $[a, b]$, is 0.12 s. The computation time for an S-IEM wave function for $[0 \leq r \leq 100]$, not including the time to interpolate to a set of equidistant mesh points, is $\simeq 0.2$ s.

8. Summary and conclusions

The present results complement the research described in Ref. [15], by showing that Milne’s Phase Amplitude method (Ph-A), combined with the spectral expansion method into Chebyshev polynomials, can be extended to regions where the potential is larger than the incident energy, i.e., in the classically forbidden region where $V - E > 0$. In this region the wave function is non-oscillatory, but it is still possible to define an amplitude $y(r)$ and a phase $\Phi(r)$ and with these obtain two linearly independent solutions of the (one-dimensional) Schrödinger equation. The connection of the wave function within the barrier region to the wave function to the right or left of the turning points of the barrier is achieved by numerically calculating the wave function by conventional methods across the turning points, and then connecting the Ph-A solution in the barrier region to these functions by a matching procedure that, incidentally, does not require knowledge of the derivative of the wave functions. The iterative Ph-A formalism converges within a radial region that is

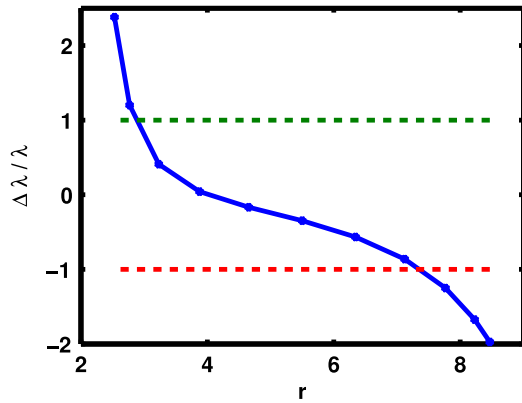


Fig. 24. For the WKB approximation to be valid, the absolute value of quantity $\Delta\lambda/\lambda$ should be smaller than 1. This is the case for $3 \leq r \leq 7$. The $\tilde{D}$ -iterations do indeed converge in this radial interval, as shown in Fig. 25. The barrier is defined with the Morse potential (22), (23) with $k = 1.0855$.

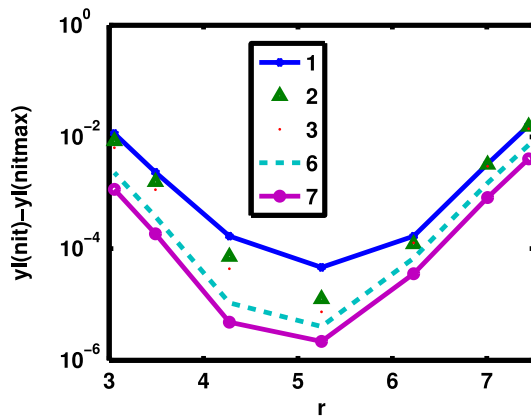


Fig. 25. Convergence of the $\tilde{D}$ -iterations in the radial interval $[3, 7.5]$. The barrier is defined with the Morse potential (22), (23) with $k = 1.0855$. For this energy k^2 the turning points occur at $r = 2$ and 12 . The calculation is based on Eq. (15), but the results based on Eq. (18) are very much the same.

contained within the region of validity of the WKB approximation, and the results are very close to – but more accurate than – the latter. This is demonstrated numerically for the case of a Morse potential that has an attractive valley near the origin, and a repulsive barrier for larger distances, and also for a repulsive Coulomb potential. One advantage of the Ph-A representation is that it provides reliable results in a shape-resonance region, for those particular energies where the wave function decreases as a function of distance inside of the barrier, especially if the barrier is very extensive and the wave function decreases by several orders of magnitude. It is for these energies where conventional methods (finite difference of finite element) solutions of the Schrödinger equation are prone to accumulate numerical errors, as was shown in previous investigations [17], [25] since the numerical errors increase (exponentially) with distance. However, for incident energies away from resonance, where the wave function increases (exponentially) in the barrier region, it is found that conventional methods of calculating the wave function are adequate.

So, in conclusion, The Ph-A method improves the accuracy of the WKB approximation in the regions where $V - E > 0$, especially in the resonance regions, as is shown with numerical

examples given for a Morse and a Coulomb potential. Furthermore, the conditions for the validity of the WKB approximation are close to identical to the conditions that assure the convergence of the iterations described in this study. Thus, too close to the turning points the iteration method will not converge.

Appendix. Convergence considerations

The conditions of the convergence of the iterations for the quantity $\tilde{D}$, described in Section 3 are linked to the convergence criterion of the WKB approximation, as will now be shown. Since the initial step of the iterations are based on the WKB approximation, this result is not surprising. The WKB conditions require that $|\Delta\lambda/\lambda| < 1$. A plot of $\Delta\lambda/\lambda$, given by Eq. (11) is shown in Fig. 24 for scattering from the Morse potential given by Eqs. (22) and (23) for the resonance case with $k = 1.0855$. According to this figure, the radial interval $3 \leq r \leq 7$ lies within the convergence of the WKB approximation, while for the domain $2.5 \leq r \leq 8$ convergence is not expected. In that radial interval convergence of the $\tilde{D}$ -iterations does indeed not take place. The iterations based on Eq. (15) for the radial interval $3 \leq r \leq 7.5$ and $k = 1.0855$ do converge, as illustrated in Fig. 25, although the accuracy after 8 iterations is only of the order of $\sim 10^{-2}$ at the edges of the radial interval. This is in contrast for the radial interval $3.5 \leq r \leq 5.5$, displayed in Fig. 6, for which the accuracy is better than 10^{-4} at the edges of that radial interval. If the $\tilde{D}$ -iterations based on Eq. (15) converge, then the iterations based on Eq. (18) will converge. But the reverse is not necessarily true, hence working only with Eq. (18) can be misleading.

So, in summary, the criteria for the validity of the WKB approximation provide a reliable guidance for the convergence of the Phase–Amplitude iterations, based on either Eqs. (15) or (18).

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